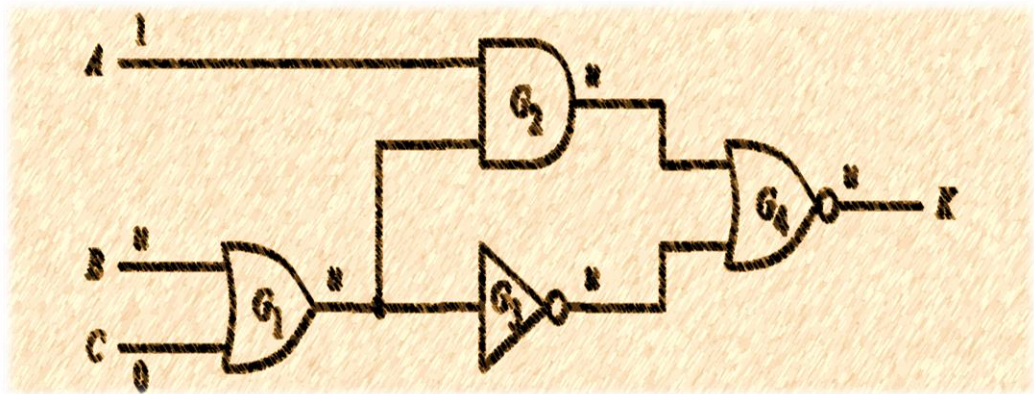


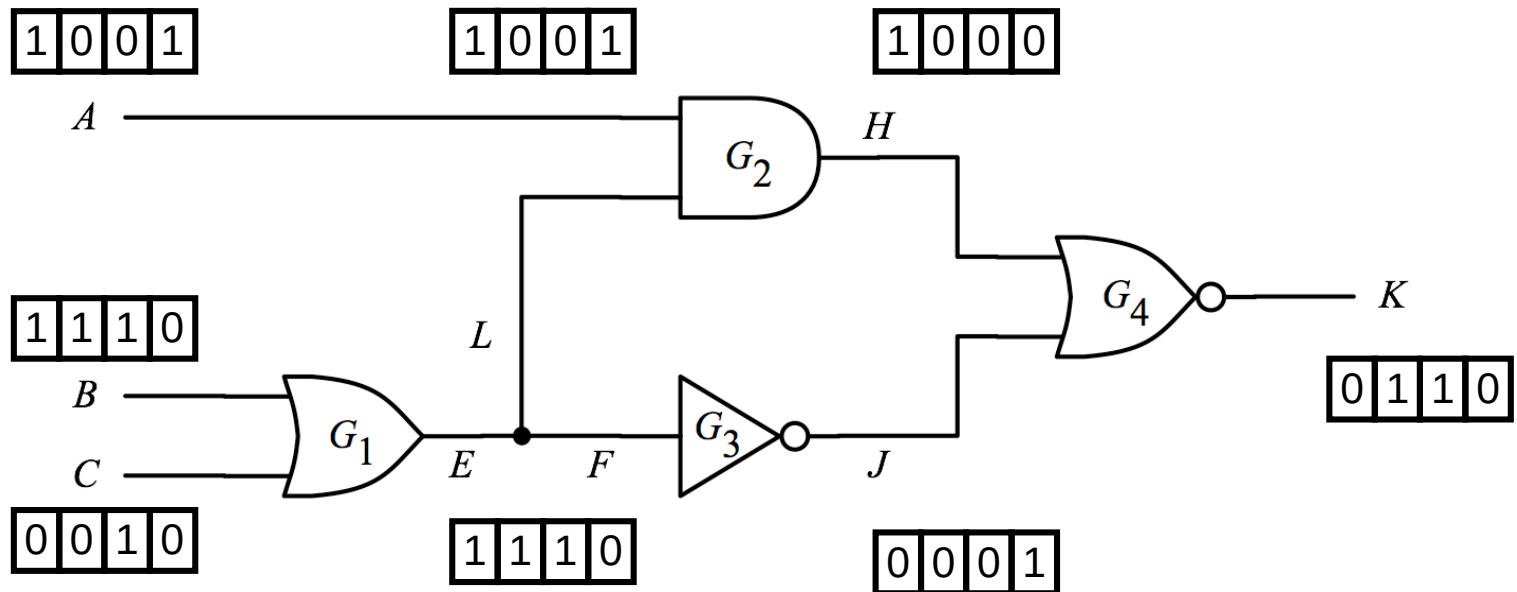
Logic Simulation

- Introduction
- Simulation Models
- Logic Simulation Techniques
 - ♦ Compiled code simulation
 - ♦ Event driven simulation
 - ♦ Parallel Simulation
- Issues of Logic Simulations
- Conclusions



Parallel Simulation

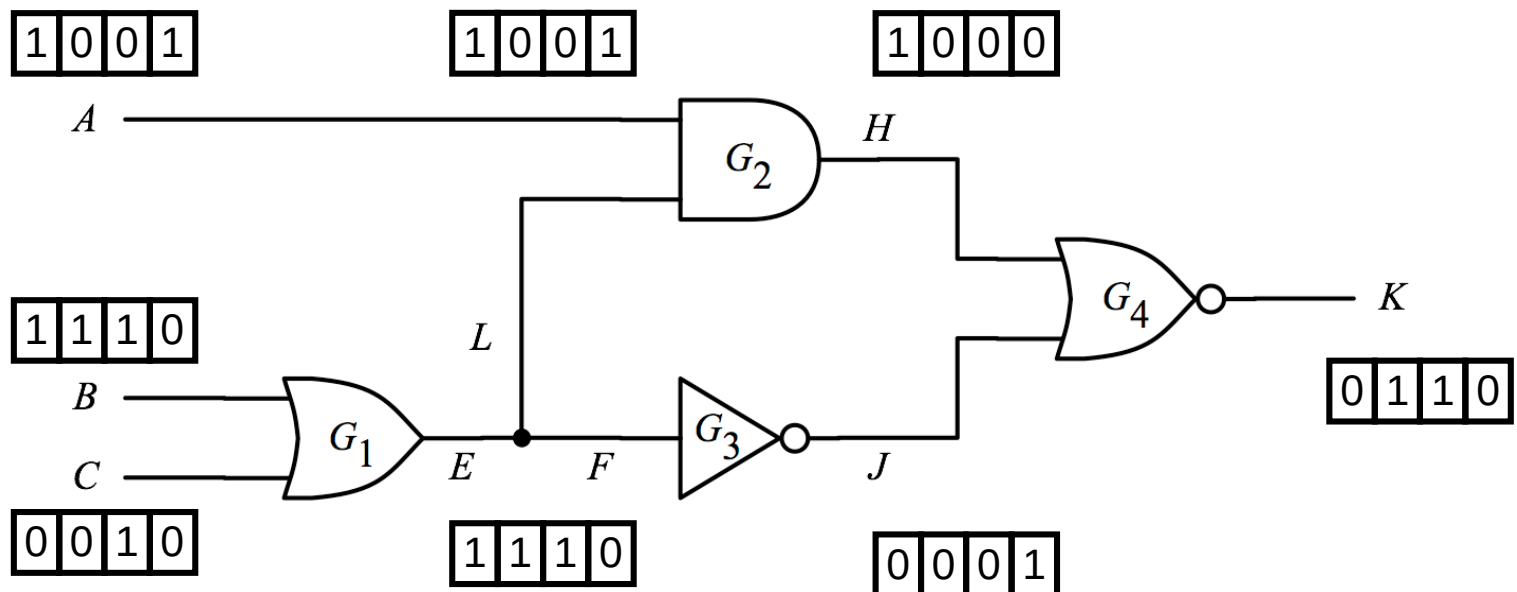
- Pack W patterns into a single word (W = word size)
 - ♦ Simultaneously evaluate a gate with W patterns
- Exploit parallelism of **bit-wise logic operation**
- Example: $W = 4$ (WWW Fig 3.8)
 - ♦ Consider only binary logic (no u or z)



Quiz

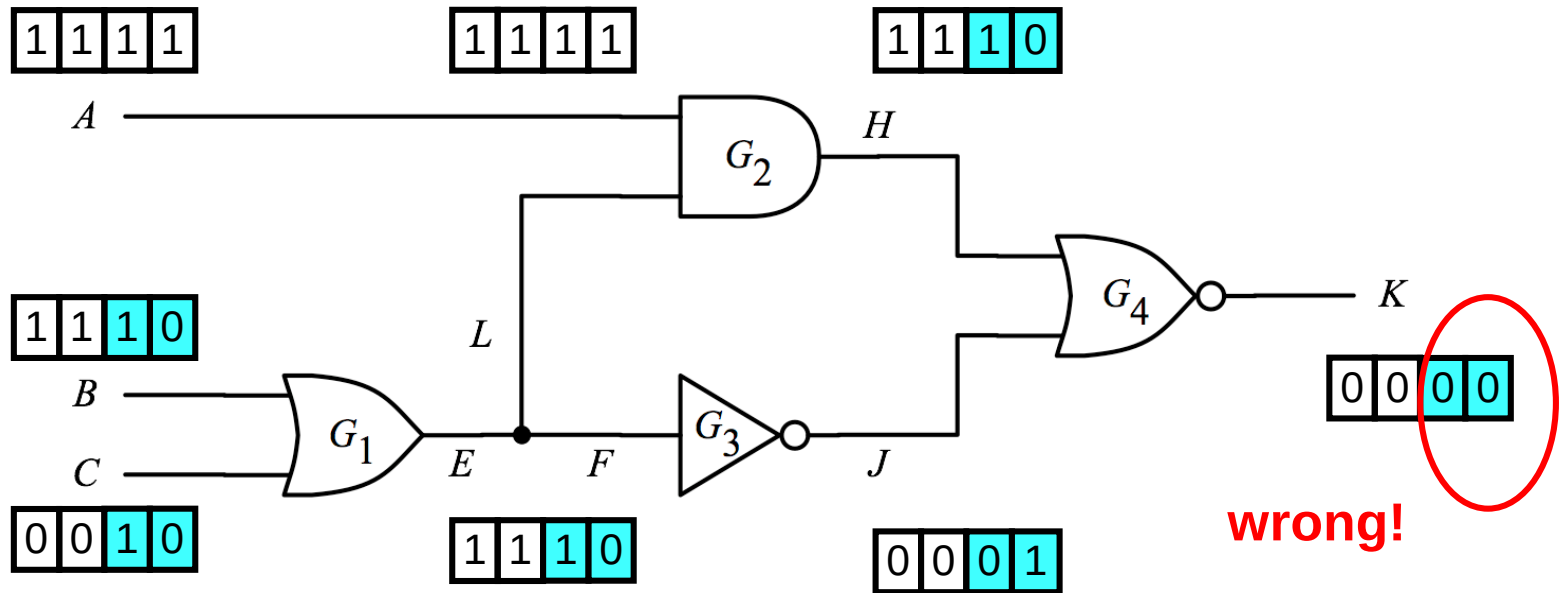
Q: what is the speedup of parallel simulation? (W = CPU size)

- A. W
- B. W^2
- C. No speedup



How about Ternary Logic?

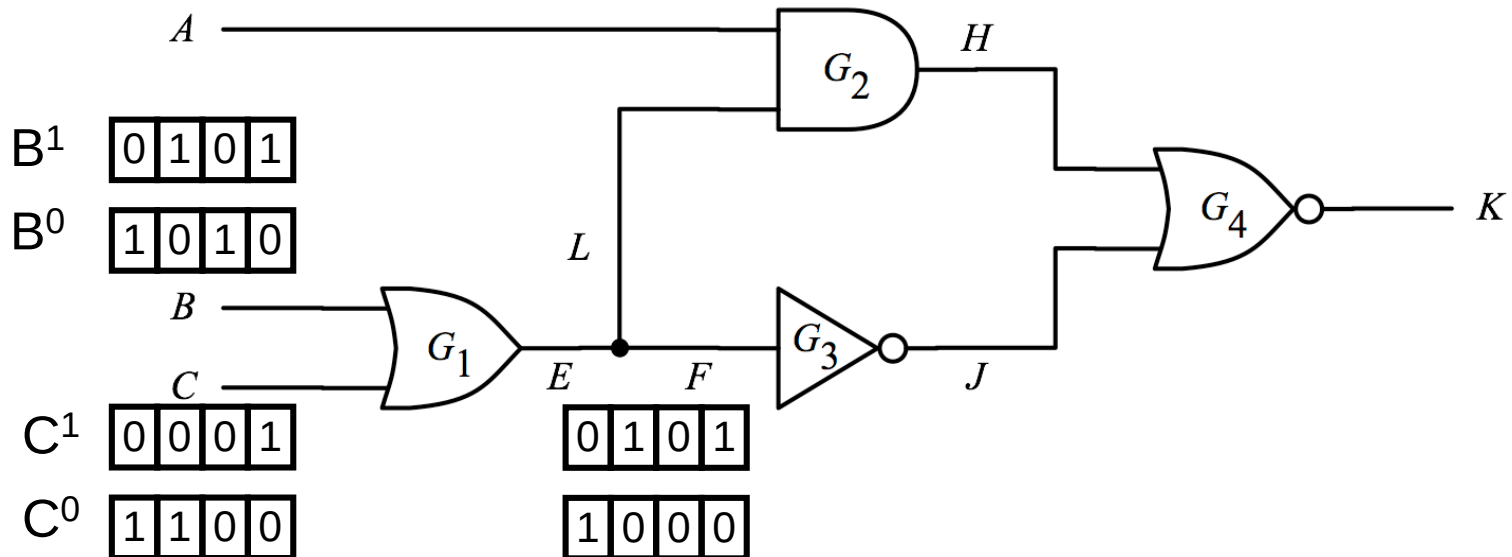
- Simple idea: use **2 bits** to represent ternary logic
- Simple encoding method
 - ♦ Logic one=11, Logic zero=00, **Unknown =10**
- Works fine with OR/AND but wrong with inversion



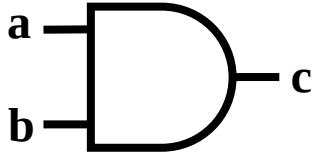
- Can we use both 10 and 01 as *u*? **No**

Sol: Improved Encoding Method

- Two words to encode single A : A^1, A^0
 - ♦ $A^1=1$ means logic one. $A^0=1$ means logic zero.
 - ♦ $A^1=0, A^0=0$ means unknown.
- Example $W=4$, four patterns
 - * $C = \{0, 0, u, 1\}$
 - * $C^1 = (0, 0, 0, 1)$
 - * $C^0 = (1, 1, 0, 0)$



Parallel Gate Evaluation



Gate	Bitwise Operations
AND	$c^1 = a^1.b^1$ $c^0 = a^0 + b^0$
NAND	$c^1 = a^0 + b^0$ $c^0 = a^1.b^1$
OR	$c^1 = a^1 + b^1$ $c^0 = a^0.b^0$
NOR	$c^1 = a^0.b^0$ $c^0 = a^1 + b^1$
INV	$c^1 = a^0$ $c^0 = a^1$

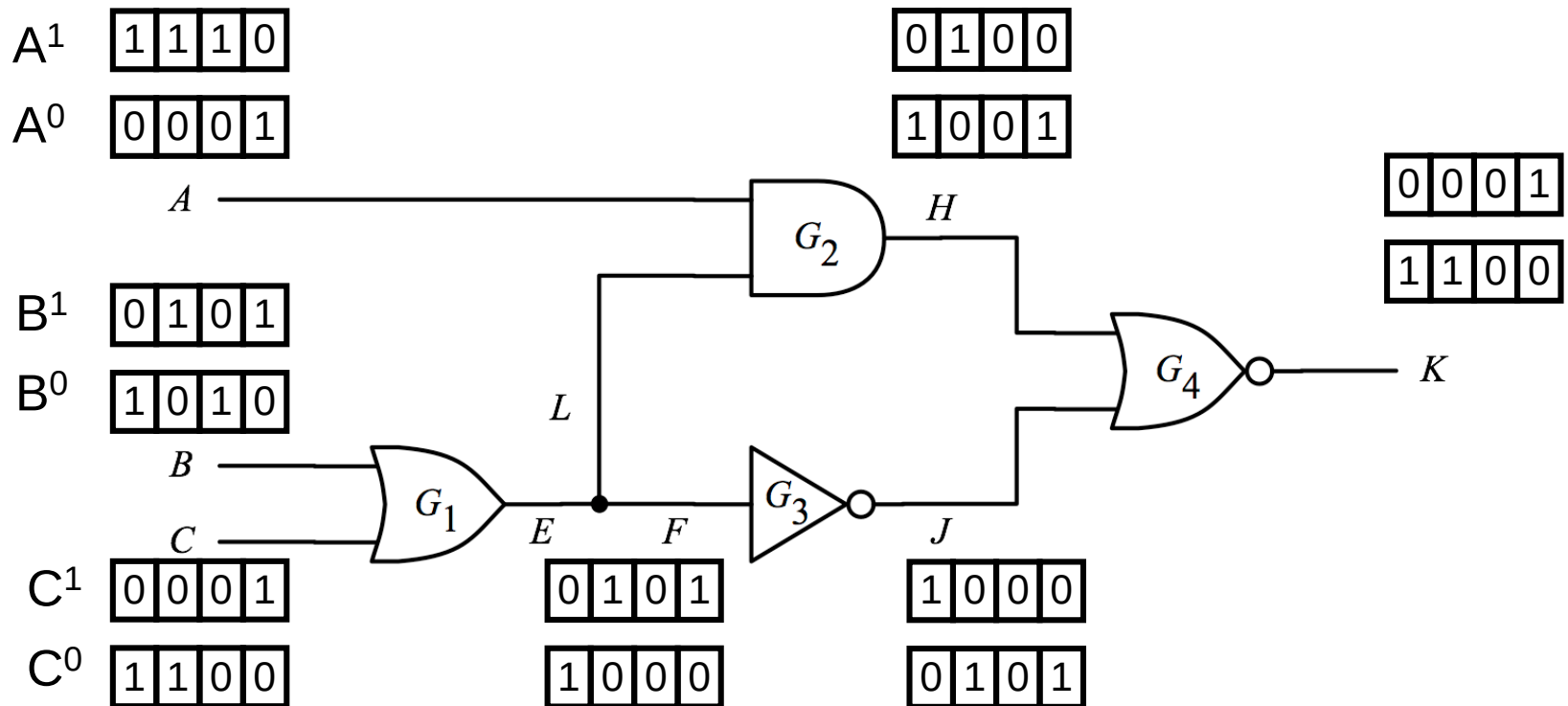
$\cdot$ = bitwise AND
 $+$ = bitwise OR

Quiz: what are equations for XOR?

A:

Example

- Apply four patterns
 - ♦ $A=\{1,1,1,0\}$; $B=\{0,1,0,1\}$; $C=\{0,0,u,1\}$
 - ♦ $K=\{0,0,u,1\}$



What is Complexity of LogicSim?

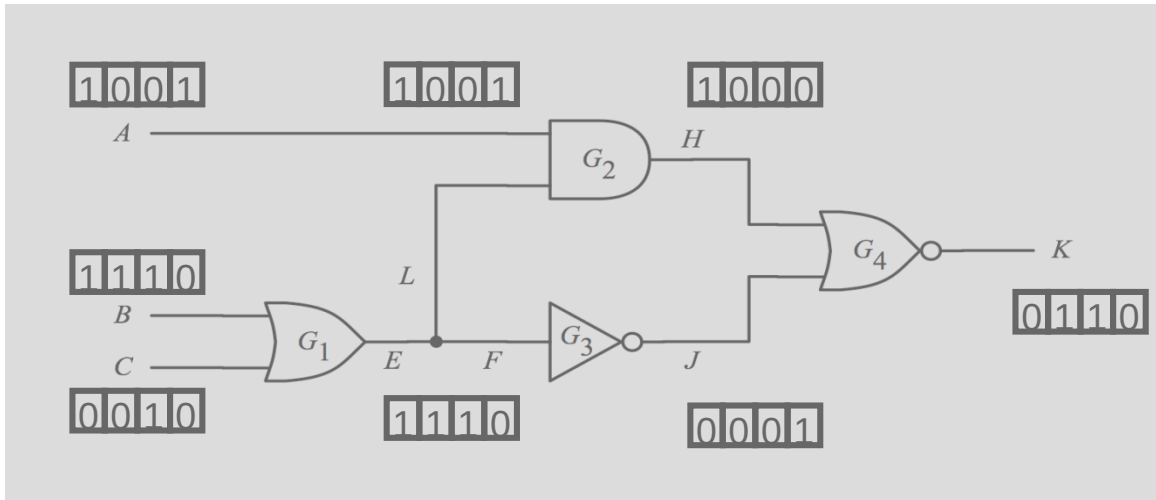
- Suppose P patterns, G gates
- Compiled-code, parallel simulation = $\Theta(P \times G)$
- Event-driven simulation = $\Theta(P \times E)$
 - ♦ E : number of events in each pattern
 - ♦ Assume $E = O(G)$
 - * $O(P \times G)$
- Logic simulation is polynomial time complexity

Logic Simulation is Polynomial Time

Summary

- Introduction
- Simulation Models
- Logic Simulation Techniques
 - ◆ Compiled code simulation
 - ◆ Event driven simulation
 - ◆ Parallel Simulation
 - * Exploits bitwise operation to gain linear speed up
 - * Improved encoding for unknowns
 - * Logic simulation is polynomial time
- Issues of Logic Simulations
- Conclusions

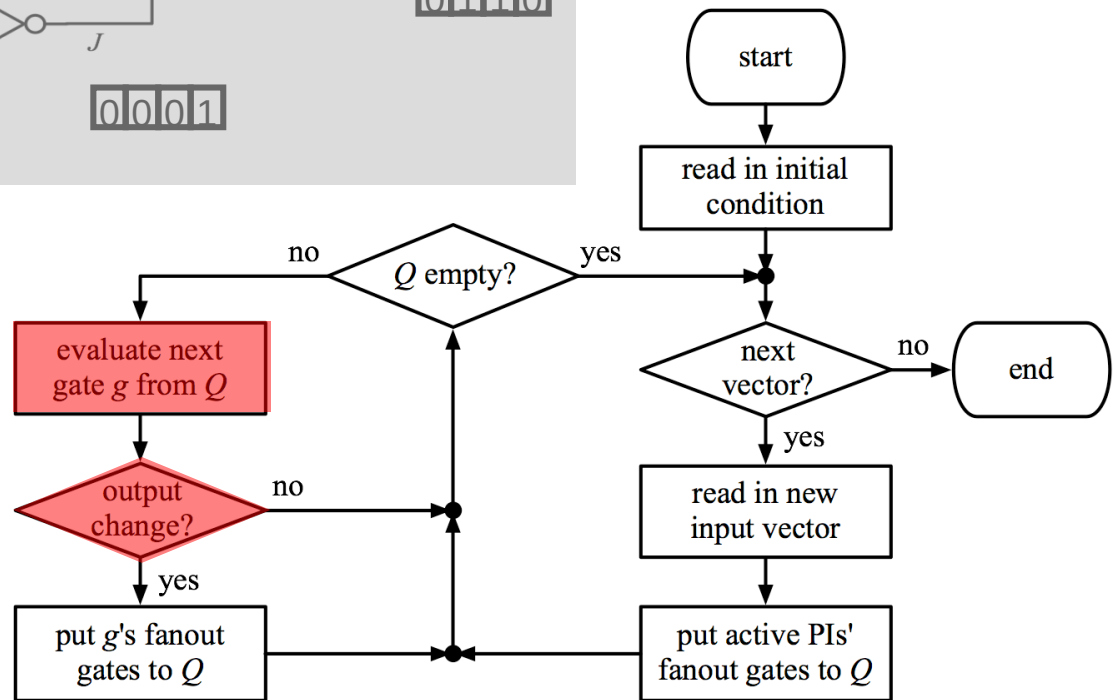
Parallel Version of Compiled-code/Event-driven



```

while{true} do
  read(A,B,C);
  E OR(B,C);
  H AND(A,E);
  J NOT(E);
  K NOR(H,J);
end
    
```

compiled-code



one-pass event-driven
(zero-delay)

FFT

- Q1: Can we swap bit pairs after inverter?
- Q2: If we can, what are advantages/disadvantages of 1-word encoding method compared with the 2-word encoding method ?

